Supplement to "What Prices Reveal About Hidden Trader Populations, and What Identifying Them Costs": a spectral blind spot and the measurement of reduced-form feedback

These two analyses accompany the paper but are not needed for its main results.

S1 A spectral blind spot

Even without identification, could a supervisor monitor risk through eigenvalues or volatility?

Theorem S1.1 (Spectral blindness)**.** *For any eigenvalues* $0 < \lambda_2 < \lambda_1 < 1$*, one-step price variance* $s_1 > 0$ *and stationary price variance* V *large enough, there is a family of two-state markets* $J_\theta = \left(\begin{smallmatrix} \lambda_1 & \theta \\ 0 & \lambda_2 \end{smallmatrix}\right)$ *with shock covariance* $\mathrm{diag}(s_1, s_2(\theta))$ *such that all members share the eigenvalues, the one-step price variance and the stationary price variance, while the worst-case transient gain* $\sup_k \|e_1^\top J_\theta^k\|$ *grows without bound in* θ*.*

In words. Two markets can look identical by the usual risk measures, the same eigenvalues and the same price variance, and yet one of them can produce far larger losses in the short run. Monitoring eigenvalues or volatility alone therefore cannot certify that a market is safe.

At $\lambda = (0.8, 0.7)$, $s_1 = 1$ and $V = 5$, the gain is 1.77, 5.10, 16.95, 50.85 and 169.50 for $\theta = 1, 3, 10, 30, 100$, with stationary variance 5.000 in every case; the cumulative gain rises from 18.3 to 1,667.7. No monitor that depends only on eigenvalues and second moments of the price can be uniformly safe (Figure S1).

Two clarifications prevent over-reading. First, the cohort economy of Section 2 is itself non-normal, yet its price output does not amplify (Lemma 2.3). Large weighted gains can be manufactured by output weights alone: with $C_w = (1, 160)$, a price-and-memory functional peaks at 160.003 at $t = 0$, which is the static weight, not dynamic growth (Appendix C.5). Second, in a network of M identical cohort economies coupled by symmetric nonnegative cross-impact, the coupled operator stays symmetric in the coordinates of

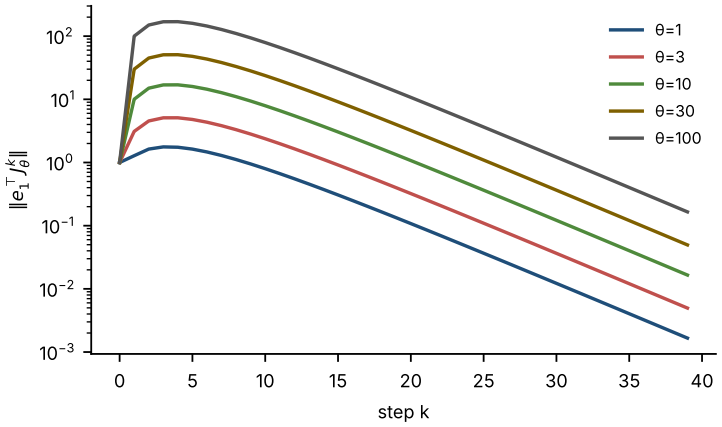


Figure S1: Price-row transient gain of the spectral-blindness family J_θ: eigenvalues $(0.8, 0.7)$, one-step price variance 1 and stationary price variance 5 are the same for every θ.

Lemma 2.3; with cross-impact 0.05, the spectral radius rises from 0.822 ($M = 1$) to 0.973 ($M = 10$) while the Kreiss constant stays at one, as it must for a stable symmetric operator (`recompute_oct20.py`). A stable normal contraction cannot exhibit Euclidean state-norm transient growth. This does not exclude scalar input-output overshoot relative to initial impact, which also depends on the input and output maps. The positive-cohort no-overshoot result is model-specific; the companion paper studies a depth-depletion mechanism.

S2 Empirical feedback and its measurement

The theory says what prices can reveal. The empirical work has two roles, and neither is a test of the identification bounds themselves, which concern controlled experiments no outside researcher can run. First, Section 6.4 tests the theory's policy prediction directly: that class-tagged data pin down memories that prices leave unrestricted. Second, this section asks what the reduced-form feedback indices a supervisor would compute from public data actually measure, and how much of them is feedback rather than filtering, timing or trading activity.

S2.1 Estimand and filtering mechanics

In theory $x_t = p_t - p_t^*$ is the deviation from an unobserved fundamental. Empirical work replaces p_t^* by a trailing filter. Let p_b be a log price proxy on uniform bars of length Δ and $m_b = (1-\alpha)m_{b-1} + \alpha p_b$ an exponential moving average with $\alpha = 2/(W/\Delta + 1)$. The empirical state is

$$x_b^{\mathrm{EMA}} \equiv p_b - m_b = (1-\alpha)(p_b - m_{b-1}) = \frac{1-\alpha}{1-(1-\alpha)L}\Delta p_b. \tag{S1}$$

Because $x_{b+1}^{\mathrm{EMA}} = (1-\alpha)(p_{b+1} - m_b)$, a martingale price gives

$$\mathbb{E}[x_{b+1}^{\mathrm{EMA}} \mid \mathcal{F}_b] = (1-\alpha)x_b^{\mathrm{EMA}} : \tag{S2}$$

the filtered state is an AR(1) with mechanical persistence $a_0 = 1-\alpha$, which equals $29/31 \approx 0.9355$ for $W = 300$ seconds and $\Delta = 10$ seconds.

Let f_t be signed flow. The empirical VAR(1)

$$\begin{pmatrix} x_{t+1}^{\mathrm{EMA}} \\ f_{t+1} \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix} \begin{pmatrix} x_t^{\mathrm{EMA}} \\ f_t \end{pmatrix} + e_{t+1} \tag{S3}$$

has reduced-form coefficients, not authenticated impact or learner parameters. When denominators are nonzero we use the scale-invariant indices

$$\hat{Q} = \frac{\hat{J}_{12}\hat{J}_{21}}{1-\hat{J}_{22}}, \qquad \hat{S} = \hat{J}_{11} + \hat{Q}, \qquad \hat{L} = \frac{\hat{Q}}{1-\hat{J}_{11}}. \tag{S4}$$

Equation (S2) implies that $\hat{J}_{11}$ must be compared with $1-\alpha$, not with zero: persistence that a naive reading attributes to feedback can be pure filtering.

Table S1: Samples

Sample	Coverage	Size	Source
Binance spot panel, 8 pairs	96 dates, 2021–2024	66.3M one-second bars	Binance archive
Binance spot fee-event windows, 8 pairs	±20 days around 2022-07-08 and 2023-03-22	640 pair-days	Binance archive
Binance USD-M perpetuals, 30 symbols	90 dates, 2023–2026	2,565 symbol-days	Binance archive
Binance BTCUSDT one-second panel	72 dates, 2021–2023	6.2M one-second bars	Binance archive
Binance BTCUSDT noise-calibration days	January 10–20, 2023	11 days	Binance archive

Binance files are verified against the published SHA-256 checksums (every file used). Sampled dates are the 10th and 24th of each month.

Table S2: Multi-year Binance spot panel: reduced-form indices under three timing constructions

Construction	$\hat{Q}$	Date interval	$\hat{S}$	Date interval	$\hat{Q} > 0$
Forward-paired flow (look-ahead)	0.0081424	[0.0072965, 0.0089882]	0.92074	[0.91665, 0.92482]	87.4%
Observed-flow EMA	0.0003733	[0.0002351, 0.0005116]	0.92072	[0.91640, 0.92504]	62.4%
Observed, 30 flow lags	0.0002580	[0.0001415, 0.0003745]	0.91738	[0.91167, 0.92308]	59.1%
Observed raw return	−0.0001555	[−0.0003732, 0.0000622]	−0.04051	[−0.05078, −0.03025]	49.2%
Placebo random walk	0.0000016	[−0.0000020, 0.0000052]	0.93510	[0.93485, 0.93534]	51.0%

768 study units (96 dates × 8 pairs), 6,634,752 ten-second transitions; equal-weight asset means within date, then date means with Student intervals. Mechanical persistence $29/31 = 0.93548$.

S2.2 Data

S2.3 Clock alignment dominates reduced-form feedback

We estimate (S3) on ten-second bars built from one-second Binance klines across a 96-date, eight-pair spot panel (the 10th and 24th of each month, 2021–2024) under three constructions (Table S2), and run the identical pipeline on a placebo random walk with each pair-day's volatility. Pairing the state with *next-bar* flow (a look-ahead) gives $\hat{Q} = 0.0081$; aligning flow with the bar in which it is observed lowers this by more than an order of magnitude, to 0.00037; and replacing the EMA state by raw returns gives an interval that includes zero. Mean $\hat{J}_{11}$ under observed flow is 0.9203, below the mechanical 0.9355, and the placebo reproduces the mechanical value (0.9351) with $\hat{Q}$ indistinguishable from zero.

Table S3: Asset means under forward-paired and observed flow

Pair	Forward-paired $\hat{Q}$	Observed $\hat{Q}$	$\hat{Q}_{30}$	Observed $\hat{S}$
BTC/USDT	0.0141870	0.0006073	0.0004891	0.93464
ETH/USDT	0.0130885	0.0005675	0.0002549	0.93511
SOL/USDT	0.0087763	0.0002129	0.0001100	0.91966
ADA/USDT	0.0067465	0.0003928	0.0003546	0.91452
DOGE/USDT	0.0058956	0.0002464	0.0001432	0.92496
BNB/USDT	0.0057420	0.0004479	0.0002762	0.89533
XRP/USDT	0.0057034	0.0002249	0.0001508	0.92362
LTC/USDT	0.0049997	0.0002870	0.0002853	0.91792

96 dates per pair. $\hat{Q}_{30}$ controls 30 ten-second bars of lagged flow. Placebo $|\hat{Q}| \leq 1.1 \times 10^{-5}$ for every pair.

Table S4: Feedback beyond filtering mechanics (Binance BTCUSDT, 72 days, 2021–2023)

	Real prices	Placebo (independent random walk)
Mean loop gain $\hat{Q}$	0.0045	0.0000009
95% CI (day bootstrap)	[0.0034, 0.0056]	[−0.0000006, 0.0000023]
Mean price persistence $\hat{J}_{11}$	0.9364	0.9357
Mechanical value $1 - 2/31$		0.9355

Placebo 95% band across days: $[-0.0000095, 0.0000156]$. This construction pairs the state with next-second flow; it is not comparable in level with the observed-flow ten-second index of Table S2.

S2.4 Feedback beyond filtering mechanics

We re-estimate feedback on a separate sample and add a placebo. We use Binance BTCUSDT one-second klines on 72 sampled days in 2021–2023. The state x_t is the log price deviation from its EMA (span 30 one-second bins), f_t is next-second signed taker flow, and a VAR(1) gives $\hat{Q}$. Because the filter alone produces persistence, we run the identical pipeline on a placebo: a random walk with the same one-second volatility, independent of the real flow.

Table S4 gives two findings. Feedback, in the sense of a loop gain the filter cannot produce, is detectable: the real $\hat{Q}$ lies far outside the placebo band. It is also small, and the price persistence that a naive reading would attribute to feedback is almost entirely mechanical (0.9364 against 0.9355). Because this construction pairs the state with next-second flow, part of $\hat{Q}$ reflects the timing effect of Table S2; the placebo shows only that the filter does not produce it.

Table S5: Feedback estimators under order splitting (simulation, 500 replications)

Estimator	Coverage at $\gamma = 0.2$	Rejection at $\gamma = 0$	Bias	RMSE
OLS	0.000	1.000	0.892	0.893
Cochrane–Orcutt, true ρ	0.944	0.054	0.001	0.039
Cochrane–Orcutt, estimated ρ	0.916	0.080	0.002	0.043
2SLS	0.938	0.044	−0.002	0.116
LIML	0.938	0.044	−0.003	0.117

Newey–West standard errors (10 lags), $T = 3{,}000$. Median first-stage $F = 224$; a placebo instrument (future z) has $F = 0.62$. Estimators outside 93–97% coverage give invalid inference.

S2.5 Order splitting and the estimation of feedback

Persistent order flow makes lagged-flow regressions inconsistent. Suppose flow follows $q_t = \gamma y_t + u_t$ with memory $y_t = (1-\eta)y_{t-1} + \eta\Delta p_{t-1}$, order splitting $u_t = \rho u_{t-1} + \nu_t$, and $\Delta p_t = \kappa q_t + \lambda z_t + e_t$ with an observed exogenous price shock z_t. Lagged returns are correlated with u_{t-1}, so OLS of flow on returns is biased even when $\gamma = 0$. Quasi-differencing ($q_t - \rho q_{t-1} = \gamma(y_t - \rho y_{t-1}) + \nu_t$) or instruments that move y through prices but are independent of u ($z_{t-1}, z_{t-2}, z_{t-3}$) restore consistency. We compare estimators over 500 replications with $\eta = 0.3$ and $\rho = 0.8$ (Table S5).

OLS is badly biased. Cochrane–Orcutt with the true ρ, 2SLS and LIML have correct size and coverage; Cochrane–Orcutt with an estimated ρ under-covers (Cochrane and Orcutt, 1949; Staiger and Stock, 1997). These are properties of estimators in a simulated market. On real data, a 2024 sub-panel of 192 spot units shows why the correction matters little at our bar length: with one VAR lag, the flow-equation residuals have essentially no first-order serial correlation (median $\hat{\rho} = -0.0002$), and Cochrane–Orcutt quasi-differencing moves the mean loop gain from 3.15×10^{-4} to 3.21×10^{-4} (post hoc). Order splitting shows up instead at longer lags, which the 30-lag control of Table S3 addresses. A real-data instrumental-variables estimate needs an instrument whose exclusion can be defended, which we do not claim to have; funding-settlement times, for example, may affect order flow directly.

Controlling for 30 ten-second bars of lagged flow lowers $\hat{Q}$ further (Table S3, column $\hat{Q}_{30}$), consistent with persistent order splitting.

Table S6: Zero fees on BTCUSDT: activity and measured feedback

	Zero fees introduced (2022-07-08)	Zero fees removed (2023-03-22)	Pooled
Log quote volume, $\hat{\beta}$	0.945 (0.074)	1.599 (0.080)	1.272 (0.084)
Observed-flow $\hat{Q}$, $\hat{\beta}$	0.00058 (0.00032)	−0.00091 (0.00060)	−0.00016 (0.00036)
p-value	0.078	0.140	0.650
Pair-days	320	320	640

$\hat{\beta}$ is the effect of zero fees on BTCUSDT relative to seven control pairs; date-clustered standard errors in parentheses. For the second event ZeroFee equals one before March 22, 2023.

S2.6 Feedback across 30 perpetual markets

The companion paper estimates the same loop gain on one-minute bars of 30 Binance USD-M perpetuals spread across the liquidity ranking (2023–2024 training days). Measured feedback rises with depth: its Spearman correlation with log depth within 1% of the mid is +0.70 [0.41, 0.87], from about 0.012 in BTCUSDT and ETHUSDT to 0.0016 on average in the thinnest third, and the placebo pipeline gives $\hat{Q} \approx 0$ for every symbol. Feedback in this reduced-form sense is a feature of deep, algorithmically traded books.

S2.7 Two fee events

Binance set spot trading fees on BTC pairs to zero on July 8, 2022 and reinstated them on March 22, 2023. Each event changes the cost of trading BTCUSDT while leaving the seven other pairs unchanged, and the second reverses the first. For every pair and every day within 20 days of each event we compute daily quote volume and the observed-flow $\hat{Q}$, and estimate $Y_{it} = \alpha_i + \gamma_t + \beta\,\mathrm{ZeroFee}_{it} + e_{it}$ with pair and date fixed effects and date-clustered errors, separately for each event and pooled (Table S6).

The fee change moved activity enormously: zero fees multiplied BTCUSDT volume relative to the controls by $e^{0.945} \approx 2.6$ when they began, and their removal divided it by $e^{1.599} \approx 4.9$. Measured feedback did not follow. The two point estimates have opposite signs, neither is significant, and the pooled effect is −0.00016 with an interval of $[-0.00087, 0.00055]$, against a sample mean of 0.00037. A strong first stage with no second-

stage response says that $\hat{Q}$ is not a by-product of trading activity, and the reversal design removes the main weakness of a single treated event. With one treated asset the interval remains wide, so this is evidence that $\hat{Q}$ does not scale with activity, not a proof of invariance.

References

Alfonsi, A., A. Schied, and A. Slynko (2012). Order book resilience, price manipulation, and the positive portfolio problem. *SIAM Journal on Financial Mathematics* 3. doi:10.1137/110822098.

Alfonsi, A., A. Schied, and F. Klöck (2016). Multivariate transient price impact and matrix-valued positive definite functions. *Mathematics of Operations Research* 41, 914–934.

Almgren, R., and N. Chriss (2001). Optimal execution of portfolio transactions. *Journal of Risk* 3, 5–39.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2015). X-CAPM: An extrapolative capital asset pricing model. *Journal of Financial Economics* 115, 1–24.

Barberis, N., R. Greenwood, L. Jin, and A. Shleifer (2018). Extrapolation and bubbles. *Journal of Financial Economics* 129, 203–227.

Biais, B., T. Foucault, and S. Moinas (2015). Equilibrium fast trading. *Journal of Financial Economics* 116, 292–313.

Crouzet, N., I. Dew-Becker, and C. G. Nathanson (2020). On the effects of restricting short-term investment. *Review of Financial Studies* 33, 1–43.

Cespa, G., and X. Vives (2015). The beauty contest and short-term trading. *Journal of Finance* 70, 2099–2154.

Amilon, H. (2008). Estimation of an adaptive stock market model with heterogeneous agents. *Journal of Empirical Finance* 15, 342–362.

Batenkov, D., G. Goldman, and Y. Yomdin (2021). Super-resolution of near-colliding point sources. *Information and Inference* 10, 515–572.

Bernstein, S. (1914). Sur la meilleure approximation de $|x|$ par des polynômes de degrés donnés. *Acta Mathematica* 37, 1–57.

Bessembinder, H., W. Maxwell, and K. Venkataraman (2006). Market transparency, liquidity externalities, and institutional trading costs in corporate bonds. *Journal of Financial Economics* 82, 251–288.

Boehmer, E., G. Saar, and L. Yu (2005). Lifting the veil: An analysis of pre-trade transparency at the NYSE. *Journal of Finance* 60, 783–815.

Bouchaud, J.-P., Y. Gefen, M. Potters, and M. Wyart (2004). Fluctuations and response in financial markets: The subtle nature of "random" price changes. *Quantitative Finance* 4, 176–190.

Bouchaud, J.-P., J. Bonart, J. Donier, and M. Gould (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.

Brock, W. A., and C. H. Hommes (1998). Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control* 22, 1235–1274.

Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27, 2267–2306.

Brunnermeier, M. K., and L. H. Pedersen (2009). Market liquidity and funding liquidity. *Review of Financial Studies* 22, 2201–2238.

Castelli, V., and T. M. Cover (1996). The relative value of labeled and unlabeled samples in pattern recognition with an unknown mixing parameter. *IEEE Transactions on Information Theory* 42, 2102–2117.

Cochrane, D., and G. H. Orcutt (1949). Application of least squares regression to relationships containing auto-correlated error terms. *Journal of the American Statistical Association* 44, 32–61.

Colliard, J.-E., T. Foucault, and S. Lovo (2022). Algorithmic pricing and liquidity in securities markets. CEPR Discussion Paper 17606.

Banerjee, S., and M. Szydlowski (2026). Trading against algorithms: Price dynamics and risk-sharing in a market with Q-learners. Working paper.

Moallemi, C. C., B. Park, and B. Van Roy (2008). Strategic execution in the presence of an uninformed arbitrageur. arXiv preprint arXiv:0801.3001.

Tremacoldi-Rossi, P. (2025). Manipulating algorithmic markets. Working paper.

Trebbi, F., M. B. Zhang, and M. Simkovic (2024). The cost of regulatory compliance in the United States. Working paper.

van Waarde, H. J., J. Eising, H. L. Trentelman, and M. K. Camlibel (2020). Data informativity: A new perspective on data-driven analysis and control. *IEEE Transactions on Automatic Control* 65, 4753–4768.

Dou, W. W., I. Goldstein, and Y. Ji (2025). AI-powered trading, algorithmic collusion, and price efficiency. NBER Working Paper 34054.

Duarte, J., and L. Young (2009). Why is PIN priced? *Journal of Financial Economics* 91, 119–138.

Easley, D., N. M. Kiefer, M. O'Hara, and J. B. Paperman (1996). Liquidity, information, and infrequently traded stocks. *Journal of Finance* 51, 1405–1436.

European Commission (2017). Commission Delegated Regulation (EU) 2017/580 of 24 June 2016 supplementing Regulation (EU) No 600/2014 with regard to regulatory technical standards for the maintenance of relevant data relating to orders in financial instruments (RTS 24), Article 2(1)(b)–(c). *Official Journal of the European Union* L 87, 31.3.2017.

European Commission (2017). Commission Delegated Regulation (EU) 2017/589 of 19 July 2016 supplementing Directive 2014/65/EU with regard to regulatory technical standards specifying the organisational requirements of investment firms engaged in algorithmic trading (RTS 6). *Official Journal of the European Union* L 87, 31.3.2017.

European Securities and Markets Authority (2016). Guidelines on transaction reporting, order record keeping and clock synchronisation under MiFID II. ESMA/2016/1452.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *Annals of Statistics* 19, 1257–1272.

Leach, J. C., and A. N. Madhavan (1993). Price experimentation and security market structure. *Review of Financial Studies* 6, 375–404.

Andrews, D. W. K., and X. Cheng (2012). Estimation and inference with weak, semi-strong, and strong identification. *Econometrica* 80, 2153–2211.

Barth, D., and R. J. Kahn (2021). Hedge funds and the Treasury cash-futures disconnect. Office of Financial Research Working Paper 21-01.

Forssell, U., and L. Ljung (1999). Closed-loop identification revisited. *Automatica* 35, 1215–1241.

Franke, R., and F. Westerhoff (2012). Structural stochastic volatility in asset pricing dynamics: Estimation and model contest. *Journal of Economic Dynamics and Control* 36, 1193–1211.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. *Quantitative Finance* 10, 749–759.

Glosten, L. R., and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14, 71–100.

Hasbrouck, J. (1991). Measuring the information content of stock trades. *Journal of Finance* 46, 179–207.

Heinrich, P., and J. Kahn (2018). Strong identifiability and optimal minimax rates for finite mixture estimation. *Annals of Statistics* 46, 2844–2870.

Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66, 1–33.

Hua, Y., and T. K. Sarkar (1990). Matrix pencil method for estimating parameters of exponentially damped/undamped sinusoids in noise. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 38, 814–824.

Katz, R., N. Diab, and D. Batenkov (2023). On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents. arXiv preprint arXiv:2302.05883.

Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The Flash Crash: High-frequency trading in an electronic market. *Journal of Finance* 72, 967–998.

Kong, W., and G. Valiant (2017). Spectrum estimation from samples. *Annals of Statistics* 45, 2218–2247.

Kreiss, H.-O. (1962). Über die Stabilitätsdefinition für Differenzengleichungen die partielle Differentialgleichungen approximieren. *BIT Numerical Mathematics* 2, 153–181.

Kukacka, J., and J. Barunik (2017). Estimation of financial agent-based models with simulated maximum likelihood. *Journal of Economic Dynamics and Control* 85, 21–45.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53, 1315–1335.

Le Cam, L. (1986). *Asymptotic Methods in Statistical Decision Theory*. Springer.

Liu, P., and H. Ammari (2022). Super-resolution of positive near-colliding point sources. arXiv

preprint arXiv:2212.00536.

Ljung, L. (1999). *System Identification: Theory for the User*, 2nd ed. Prentice Hall, Upper Saddle River, NJ.

Makarov, I., and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics* 135, 293–319.

Manole, T., and N. Ho (2022). Refined convergence rates for maximum likelihood estimation under finite mixture models. *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162.

Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer, New York.

Niroula, P., S. Alkhushayni, Y. Chen, M. Bhavsar, and P. Niroula (2026). From human markets to machine markets: Transient amplification, an exact depth condition, and a maximal pre-trade guard. Working paper, Minnesota State University, Mankato.

Moitra, A. (2015). Super-resolution, extremal functions and the condition number of Vandermonde matrices. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 821–830.

Moskowitz, T. J., Y. H. Ooi, and L. H. Pedersen (2012). Time series momentum. *Journal of Financial Economics* 104, 228–250.

Obizhaeva, A. A., and J. Wang (2013). Optimal trading strategy and supply/demand dynamics. *Journal of Financial Markets* 16, 1–32.

Pagano, M., and A. Röell (1996). Transparency and liquidity: A comparison of auction and dealer markets with informed trading. *Journal of Finance* 51, 579–611.

Park, B., and B. Van Roy (2015). Adaptive execution: Exploration and learning of price impact. *Operations Research*, doi:10.1287/opre.2015.1415.

Rebafka, T., and F. Roueff (2015). Nonparametric estimation of the mixing density using polynomials. arXiv preprint arXiv:1002.4516.

Rivlin, T. J. (1981). *An Introduction to the Approximation of Functions*. Dover.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. *Journal of Finance* 39, 1127–1139.

Roy, R., and T. Kailath (1989). ESPRIT—estimation of signal parameters via rotational invariance techniques. *IEEE Transactions on Acoustics, Speech, and Signal Processing* 37, 984–995.

Consolidated Audit Trail, LLC (2026). Consolidated Audit Trail, LLC 2026 Financial and Operating Budget, 31 March 2026 (total expenses $147,044,869). https://www.catnmsplan.com/cat-financial-and-operating-budget.

U.S. Securities and Exchange Commission (2010). Risk management controls for brokers or dealers with market access. Release No. 34-63241.

U.S. Securities and Exchange Commission (2012). Consolidated Audit Trail; Final Rule (Release No. 34-67457, July 18, 2012, 77 FR 45722 at 45790–45795, pp. 220–238; proposing release No. 34-62174, May 26, 2010, 75 FR 32556 at 32598, 32600, pp. 147, 153).

U.S. Securities and Exchange Commission (2016). Joint Industry Plan; Order Approving the National Market System Plan Governing the Consolidated Audit Trail. Release No. 34-79318, November 15, 2016, 81 FR 84696 at 84801, p. 363.

U.S. Securities and Exchange Commission (2024). Notice of Filing of Proposed Rule Change Relating to the 2025 CAT Budget. Release No. 34-101901, December 12, 2024, p. 4 (revised May 2025).

U.S. Securities and Exchange Commission (2025). Order granting exemptive relief from certain requirements of the National Market System Plan governing the Consolidated Audit Trail. Release No. 34-102386, February 10, 2025.

Shahram, M., and P. Milanfar (2005). On the resolvability of sinusoids with nearby frequencies in the presence of noise. *IEEE Transactions on Signal Processing* 53, 2579–2588.

Shleifer, A., and R. W. Vishny (1992). Liquidation values and debt capacity: A market equilibrium approach. *Journal of Finance* 47, 1343–1366.

Sornette, D., S. Lera, J. Lin, and K. Wu (2023). Non-normal interactions create socio-economic bubbles. *Communications Physics* 6, 261.

Staiger, D., and J. H. Stock (1997). Instrumental variables regression with weak instruments. *Econometrica* 65, 557–586.

Trefethen, L. N., and M. Embree (2005). *Spectra and Pseudospectra: The Behavior of Nonnormal*

Matrices and Operators. Princeton University Press.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer.

Varga, R. S., and A. J. Carpenter (1985). On the Bernstein conjecture in approximation theory. *Constructive Approximation* 1, 333–348.

Wagenmaker, A., and K. Jamieson (2020). Active learning for identification of linear dynamical systems. In *Proceedings of the 33rd Conference on Learning Theory*, PMLR 125, 3487–3582.

Wagenmaker, A. J., M. Simchowitz, and K. Jamieson (2021). Task-optimal exploration in linear dynamical systems. In *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139, 10641–10652.

Wellman, M. P. (2006). Methods for empirical game-theoretic analysis. In *Proceedings of the 21st National Conference on Artificial Intelligence (AAAI-06)*, 1552–1555.

Whittle, P. (1953). Estimation and information in stationary time series. *Arkiv för Matematik* 2, 423–434.

Wu, Y., and P. Yang (2020). Optimal estimation of Gaussian mixtures via denoised method of moments. *Annals of Statistics* 48, 1981–2007.